Experimental VLE, consistency and fitting

Session 2: Independent study with calculations · 180 minutes

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A three-hour independent lesson

Your action: Read this schedule, then advance. All times are elapsed lesson time, including your calculations.

This lesson replaces the class on **16 October 2026**.

ACTIVITY	MINUTES	ELAPSED
Preparation and goals	10	0–10
Phase and material-balance checks	25	10–35
From Raoult’s law to liquid nonideality	30	35–65
Antoine conventions	25	65–90
Break	10	90–100
Lab 08 investigation	45	100–145
Review test, corrections, exit and feedback	35	145–180

Times include reading and **working through calculations**. Pause before each solution.

Preparation · 0–10 min

Your action: Open worksheet pages 1–4 and read this slide. Do not start the exercises or Lab yet.

Download the 4-page worksheet. Keep this slide deck and the worksheet open side by side. A calculator and paper are enough until minute 100.

Last lesson: ideal-mixture VLE and phase calculations. Today: how we describe departures from the ideal-liquid prediction.

The 3-page reading is optional background. Skim it if time permits. This deck contains the instructions required for today.

By the end, you should be able to:

- Explain the new activity coefficient and calculate it from one VLE observation.
- Check an Antoine convention before using its constants.
- Compare two fitted models and state what the comparison supports.

1. Composition and phase amount · 10–15 min

Your action: Read this slide only. No worksheet answer is required yet. The next slide starts page 1, Section A.

SYMBOL	MEANING
z_i	Overall mole fraction in the feed
x_i	Mole fraction in the liquid phase
y_i	Mole fraction in the vapor phase
$\beta = V/F$	Fraction of total moles in the vapor phase

$$z_i = (1 - \beta)x_i + \beta y_i$$

For each existing phase, its mole fractions sum to one.

Pause and calculate · 15–23 min

Your action: Open worksheet page 1. Complete A1–A3 using this slide and your calculator. No Lab is needed.

Ideal binary mixture at 350 K: z_1=0.5, P_1^{sat}=150 kPa, P_2^{sat}=60 kPa.

$$P_{bubble} = \sum_i z_i P_i^{sat}, \qquad P_{dew} = \left(\sum_i \frac{z_i}{P_i^{sat}} \right)^{-1}$$

1. Calculate both boundary pressures for this overall composition.
2. Identify the phase state at 80, 95 and 110 kPa.
3. At 110 kPa, what are β and x_1? Does a vapor composition exist?

Finish A1–A3 before advancing to the solution.

Phase check · 23–27 min

Your action: Compare this solution with page 1, A1–A3. Add corrections beside your original answers.

$$P_{bubble} = 0.5(150) + 0.5(60) = 105\ \text{kPa}$$

$$P_{dew} = [0.5/150 + 0.5/60]^{-1} = 85.714\ \text{kPa}$$

PRESSURE	EQUILIBRIUM STATE
80 kPa	Vapor, below the dew boundary
95 kPa	Liquid + vapor, between the boundaries
110 kPa	Liquid, above the bubble boundary

At 110 kPa, $\beta=0$ and $x_1=z_1=0.5$. No vapor phase exists, so y is undefined.

Pause: material balance · 27–32 min

Your action: On worksheet page 1, complete A4: material balance and composition sums. Stop before the next solution.

At 95 kPa, use x_1=0.388889, y_1=0.614035 and β=0.493506.

Calculate:

$$z_1 = (1 - 0.493506)(0.388889) + 0.493506(0.614035)$$

Then find x_2 and y_2. Check each phase sum separately.

Use the numbers on this slide. No Lab is required for this calculation.

Material-balance check · 32–35 min

Your action: Check page 1, A4 against this slide. Keep the substituted balance and correct any mismatch.

$$z_1 \approx 0.196970 + 0.303030 = 0.500000$$

x_2=0.611111 and y_2=0.385965.

$$x_1 + x_2 = 1, \qquad y_1 + y_2 = 1$$

An incorrect phase sum can reveal a transcription error even when the plotted result looks plausible.

2. Starting from Raoult's law · 35–39 min

Your action: Read this slide. On scratch paper, calculate 60/48 before advancing. No new worksheet entry is required.

For an ideal liquid mixture with ideal vapor:

$$y_i P = x_i P_i^{sat}(T)$$

The liquid mole fraction scales the pure-component vapor pressure.

At x_1=0.4 and P_1^{sat}=120 kPa, this predicts a partial pressure of **48 kPa**.

Suppose the observation instead gives $y_1 P$=0.6(100)=**60 kPa**. The ideal-liquid prediction is too low.

Think: what factor would multiply 48 to give 60?

A new quantity: activity coefficient · 39–42 min

Your action: Read this definition of gamma. Compare 1.25 with your scratch calculation. No Lab or worksheet entry is required.

The factor is 60/48=**1.25**. We call it the **activity coefficient**, written γ (gamma).

$$y_i P = x_i \gamma_i P_i^{sat}(T)$$

- $\gamma_i = 1$ recovers Raoult's law for the liquid.
- $\gamma_i > 1$ gives a higher partial pressure than the ideal-liquid prediction.
- $\gamma_i < 1$ gives a lower partial pressure under the same assumptions.

We still assume ideal vapor and negligible Poynting correction. Verify the observations and vapor-pressure inputs before attributing a mismatch to liquid nonideality.

Activity and its reference · 42–45 min

Your action: Read this slide and check 1.25 × 0.4 on your calculator. Keep worksheet B blank until minute 48.

$$a_i = \gamma_i x_i$$

Activity a_i represents liquid composition adjusted for nonideal behavior. Here the reference is the **pure liquid**. Both a_i and γ_i are dimensionless.

For an ideal liquid solution, $\gamma_i=1$ and $a_i=x_i$.

For $x_1=0.4$ and $\gamma_1=1.25$, $a_1=0.50$.

Calculating gamma from VLE · 45–48 min

Your action: Read the equation and assumptions here. You will use them in worksheet page 1, B1–B3 on the next slide.

Rearrange the modified Raoult equation to estimate gamma from an observation.

With ideal vapor and negligible Poynting correction:

$$y_i P = x_i \gamma_i P_i^{sat}(T)$$

$$\gamma_i = \frac{y_i P}{x_i P_i^{sat}(T)}$$

Use the same pressure units in P and P^{sat}. The inferred γ depends on the vapor-pressure correlation and these assumptions.

Pause: one observation, two property inputs · 48–57 min

Your action: On worksheet page 1, complete B1–B3. Use only the observation on this slide. No Lab is needed.

Synthetic observation: $P=100$ kPa, $x_1=0.4$, $y_1=0.6$.

1. With $P_1^{sat}=120$ kPa, calculate γ_1 and a_1.
2. Repeat using $P_1^{sat}=150$ kPa.
3. Which input changed? Did the observed mixture data change?
4. Could fitting alone tell you which vapor pressure is correct?

Show substitution and unit cancellation in B1–B3.

A property error changes inferred activity · 57–65 min

Your action: Check page 1, B1–B3 with this slide. Correct your values and finish the explanation in B3.

ASSUMED P_1^{sat}	$\Gamma_1 = Y_1P/(X_1P_1^{sat})$	$A_1 = \Gamma_1X_1$
120 kPa	60/48 = 1.25	0.50
150 kPa	60/60 = 1.00	0.40

The observations stay unchanged. Increasing the assumed vapor pressure lowers the inferred γ.

A fitted model can absorb a property error into its parameters. Check the property source independently.

Now correct your work and write one sentence explaining the difference.

3. Antoine conventions · 65–72 min

Your action: Read this slide. Open worksheet page 2, C, ready for the next exercise. Do not fill the Lab section D yet.

A common form is:

$$\log_{10}(P^{sat}/P_{unit}) = A - \frac{B}{T_{source} + C}$$

Before entering constants, identify:

- The exact equation and log base, including the sign of C.
- The temperature and pressure units specified by the source.
- The valid temperature range and component identity.

Changing a menu setting does not automatically convert the constants.

Pause: Antoine calculation · 72–82 min

Your action: On worksheet page 2, answer C1–C4 using this synthetic correlation. Do not enter these constants in Lab 08.

A **synthetic teaching correlation**, valid from 20 to 80 °C:

$$\log_{10}(P^{sat}/\mathrm{kPa}) = 4 - \frac{1000}{T_C + 200}$$

1. Calculate P^{sat} at 50 °C.
2. Explain why entering 323.15 directly for T_C is incorrect.
3. If the left side were ln instead of $\log_{10}$, could the same constants be used unchanged?
4. Should this correlation be used at 100 °C?

Write the equation and substitutions, not only the answer.

Antoine check · 82–90 min

Your action: Check page 2, C1–C4 with this slide. Write corrections and their reasons before taking the break.

$$4 - \frac{1000}{50 + 200} = 0, \qquad P^{sat} = 10^{0}\ \mathrm{kPa} = 1\ \mathrm{kPa}$$

- The equation expects °C. Convert 323.15 K to 50 °C first.
- A change of log base requires corresponding constants. At this particular zero exponent, both bases happen to give 1, so this single point cannot detect the mistake.
- 100 °C lies outside the stated valid range.

Correct your calculation before moving on.

Break · 90–100 min

Your action: Pause for ten minutes. Keep your worksheet. At minute 100, open Lab 08 using the link below.

Take ten minutes away from the screen.

Keep your calculations. Next, you will use the same checks on the synthetic dataset in Lab 08.

Open Lab 08

4. Lab baseline · 100–110 min

Your action: Open Lab 08 and worksheet page 2, D1–D2. Follow the four steps below. Save the baseline JSON before continuing.

In Lab 08:

1. Select **Load synthetic example**.
2. Inspect the data units and both Antoine equation forms and constants.
3. Select **Validate & inspect data**. Correct input errors. Keep endpoint handling at **No extrapolation**.
4. Use **Export complete study JSON** to save your baseline.

Record the convention and valid range in D1–D2. Range fields use **kelvin**. A consistency result marked “not-assessed” or “not-applicable” is not an input error.

Pause: an independent Lab check · 110–120 min

Your action: Keep Lab 08 open. Complete page 2, D3 and page 3, E1 for component 1 at the same selected row.

Choose one interior observation from **Imported observations**.

Write its T, P, x_1 and y_1. The normalized table uses K and kPa.

1. Convert T to the unit required by the correlation and calculate P_1^{sat}.
2. Convert P_1^{sat} to kPa if needed.
3. Calculate $\gamma_1 = y_1 P/(x_1 P_1^{sat})$. The table shows **ln** γ_1. Compare your result with **exp(table ln** γ_1**)**, allowing for display rounding.

Complete D3 on page 2 and E1 on page 3.

If values disagree: check component order, units and log base first.

Model comparison · 120–135 min

Your action: In Lab 08, use Fit selected model for each model below. Complete worksheet page 3, E2 and save both model exports.

Fit **Margules · 1 parameter**, then **NRTL · fixed** α, using α=0.3.

Keep the observations and property inputs unchanged. Leave temperature-dependent parameters off for this comparison.

For each fit, select **View: Pressure and vapor residuals**. Record:

- Mean absolute pressure error (%) and vapor-composition error (percentage points).
- One residual pattern (e.g., sign or trend with x_1) and any parameter-bound warning.

Export each model's **complete study JSON** and **fitted residuals CSV** before switching models. Give the files distinct names.

Interpreting the comparison · 135–145 min

Your action: Read this table, then complete worksheet page 3, F using your E2 results. No additional Lab or dataset is required.

EVIDENCE	WHAT IT ESTABLISHES
Solver convergence	The optimizer met its stopping criterion
Small training residuals	The model reproduces the fitted observations
Consistency diagnostic	Agreement with a thermodynamic constraint under stated assumptions
Held-out observations	Evidence about prediction beyond the fitting data

Complete F on worksheet page 3: state one finding, its numerical evidence and one limitation.

The example data were generated with NRTL. Its close fit is expected model recovery, not evidence of universal superiority. Consistency and prediction require separate checks.

Review test: purpose · 145–147 min

Your action: Open worksheet page 4, Review test. Read these rules before answering. The questions appear on the next two slides.

This is **practice with attendance credit**. Credit is for completing the activity, rather than a minimum accuracy score.

Work individually. You may consult the lesson and your calculations.

Write answers to the next six questions, then review the solutions. Preserve your original answers and corrections.

The attendance submission route will be provided separately. The anonymous exit/feedback form does **not** identify attendance.

Review test: phase and activity · 147–155 min

Your action: Write Q1–Q3 in the matching spaces on worksheet page 4. Show your calculations before advancing.

1. Ideal binary mixture: $z_1=0.5$, $P_1^{sat}=160$ kPa, $P_2^{sat}=80$ kPa. At $P=130$ kPa, identify the phase and its composition. Show a boundary calculation.

2. For a separate binary mixture in two-phase equilibrium, $x_1=0.3$, $y_1=0.7$ and $z_1=0.5$. Calculate β by material balance.

3. For $P=90$ kPa, $x_1=0.3$, $y_1=0.5$ and $P_1^{sat}=100$ kPa, calculate γ_1 and a_1. Assume ideal vapor and negligible Poynting correction.

Keep your substitutions and units.

Review test: properties and evidence · 155–165 min

Your action: Write Q4–Q6 on worksheet page 4. For Q6, reuse your page 3, E2 results. Finish before viewing solutions.

4. A synthetic correlation is $\log_{10}(P^{sat}/kPa)=5-1200/(T_C+200)$, valid at 20–80 °C. Find P^{sat} at 40 °C. Can you insert 313.15 directly for T_C?

5. NRTL gives a lower training error than Margules. Does that establish thermodynamic consistency and better prediction? Explain.

6. From your Lab 08 results, report one comparison with its metric and units. State one limitation of your conclusion.

Finish your own answers before opening the next slide.

Review solutions · 165–169 min

Your action: Check worksheet page 4, Q1–Q4. Add corrections beside the original answers, using extra paper if needed.

1. P_bubble=0.5(160)+0.5(80)=120 kPa. At 130 kPa the mixture is liquid, $x_1=0.5$ and $\beta=0$.

2. $\beta=(z_1-x_1)/(y_1-x_1)=(0.5-0.3)/(0.7-0.3)=0.5$.

3. $\gamma_1=0.5(90)/[0.3(100)]=1.5$ and $a_1=1.5(0.3)=0.45$.

4. $5-1200/(40+200)=0$, so $P^{sat}=1$ kPa. Convert 313.15 K to 40 °C first.

Mark any correction beside your original answer.

Review of reasoning · 169–172 min

Your action: Check page 4, Q5–Q6. Explain each correction. Retain all four worksheet pages and the Lab exports for attendance submission.

5. Lower training error alone establishes neither claim. Consistency needs an applicable thermodynamic diagnostic. Prediction needs observations excluded from fitting.

6. A complete answer identifies the same dataset, reports both model results with the metric and units, and gives a defensible limitation.

For example: “Model ___ has lower mean absolute pressure error on this dataset. I have not tested it on excluded observations.” Fill the blank using **your own result**.

Exit ticket and feedback · 172–180 min

Your action: Open the Session 2 form using the QR code or link. Use page 3, F for reasoning, page 2, C/D for the property check, and E1 for numerical evidence.

5 min: explain and check

Does a small fitting residual establish thermodynamic consistency? Explain and name one independent check of the vapor-pressure inputs. Add one substituted γ or P^{sat} calculation.

3 min: feedback on independent learning

Report pace and Lab access. Use the comment box to describe any difficulty with this format.

The form requests no name, ID or email. Wait for **“Response received.”** Keep attendance work separately.



Session 2 exit ticket and feedback

Open Session 2 form